Module 12

Financial Engineering in Quantitative Investment and Portfolio Management

November 17 & 19, 2025

Enhancement for Portfolio Management

Hedging with Stock Index Futures:

Why hedging?

- Eliminate market risk, isolate "alpha"
- Lock-in gain for a period in a long-term holding

  → saving on tax, transaction cost

Hedging ratio:

Consider

$\Pi = \Pi_S + N_F V_F$

Π_S : portfolio of stocks

V_F : value of the future contract

N_F : number of future contracts. How many? → depends on the objectives of hedging

An example:

Select N_F to minimize variance:

$\delta\Pi = \Pi_S R_S + N_F V_F R_F$,

where R_S, R_F = relative changes of Π_S, V_F for the hedging period

→ $\sigma_\Pi^2 = \Pi_S^2 \sigma_S^2 + (N_F V_F)^2 \sigma_F^2 + 2\Pi_S N_F V_F \sigma_{SF}$

where σ_Π^2, σ_S^2, σ_F^2 are the variance of the returns in Π, Π_S, V_F, σ_{SF} is the covariance of returns in Π_S and V_F.

Minimize the variance σ_{Π}^2 with respect to N_F:

$$N_F = -\frac{\Pi_S \sigma_{SF}}{V_F \sigma_F^2}$$

If F is the SP500 index future, $R_F \approx R_M$ → $N_F = -\frac{\Pi_S}{V_F}\beta_S$

Adjusting Market Exposure:

To achieve desired market exposure from β_S to $\tilde{\beta}$, select F be market index future so that $R_F \approx R_M$, $\beta_F \approx 1$:

$$N_F = \frac{\Pi_S}{V_F}(\tilde{\beta} - \beta_S)$$

→ $\delta\Pi = \Pi_S R_S + N_F V_F R_F \approx \Pi_S R_S + \Pi_S(\tilde{\beta} - \beta_S) R_M$

Assume

$$R_S = \alpha_S + \beta_S R_M + \varepsilon_S$$

where ε_S = portfolio specific risk with zero mean and uncorrelated with the market

$$\delta\Pi \approx \Pi_S(\alpha_S + \beta_S R_M + \varepsilon_S) + \Pi_S(\tilde{\beta} - \beta_S) R_M = \Pi_S(\alpha_S + \tilde{\beta} R_M + \varepsilon_S)$$

→ Π responds to the market movement R_M in the same way as Π_S but with $\tilde{\beta}$

A special case:

$N_F = -\frac{\Pi_S}{V_F}\beta_S$ → $\tilde{\beta} = 0$ → Π is "market neutral":

- Π_S is a set of long positions of "cheap" stocks, but may slide further due to downward movements of market Π_M → short index future
- Π_S is a set of short positions of "expensive" stocks, but may clime further due to upward movements of market Π_M → long index future

→ $\Pi = \Pi_S + N_F V_F$ is free of "market risk"

Insurance:

- Protective put – may be done either with put on single names or on indices, but can be expensive if implied volatility is high
- Synthetic put with futures (Rubinstein & Leland 1981) – may be more cost effective than trading cash positions
    - dynamic replication: sell low, buy high
    - blamed for 1987 "Black Monday"

Income generations with covered option writing:

- Income generation/exit strategy: Covered (OTM) call writing
- Income generation/share acquisition: OTM put writing

Examples: ETF deployed with such strategies: JEPI, JEPQ, HELO

Extreme events hedging:

- Deep OTM options for hedging melt-up/down
- Black-Swan funds

Leveraged market exposure:

- ITM call: leveraged exposure to the underlying stock performance without paying too much time value
- Equity swaps: cost of such a position is its initial margin ("independent amount", IA) instead of the full amount of the underlying asset value. Size of IA depends on the potential future exposure (PFE, see Module 12). Equity swaps on market indices or on a bespoken basket of assets can be used for both investment and hedging as discussed in previous lectures.

Investment signals from derivatives markets

- Market sentiments signaled from option open interests
- Future market volatility forecasted from option implied volatility
- Expected return distribution deduced from option markets, e.g., skewness

Beyond mean-variance - shape of return distribution matters!

- compound annual growth rate
- average annual return rate

- Future asset correlations implied from correlation swaps
- Signals from the derivative markets in other asset classes (e.g., CDS markets)

Other Financial Engineering Strategies/Products

Pair trading using futures:

Suppose the stocks of company A, B are candidates for a long-short trading strategy. This could be due to, e.g., S_A and S_B deviate significantly from a historical stable relative pricing pattern, or S_A and S_B becomes a pair of M&A play.

One may set up portfolio

$$\Pi = N_A V_A - N_B V_B + N_I V_I$$

N_A, N_B, N_I are the number of future contracts on A, B and market index, V_A, V_B, V_I are the value of the future contracts. N_I is chosen to hedge out the residual sensitivity to the overall market movements of the long-short position so that Π is market neutral.

Constant Proportion Portfolio Insurance (CPPI):

$$CPPI(t) = w_r(t)\Pi_{risky}(t) + w_s(t)\Pi_{safe}(t)$$

Π_{risky} : portfolio of risky assets (stock, commodity, currency, etc)

$\Pi_{safe}(t)$: portfolio of safe assets (e.g., risk free bonds or other low risk assets))

Risky asset level RAL :

$$RAL(t) = m\frac{CPPI(t) - Floor(t)}{CPPI(t)} \qquad (= \% \text{ of risky asset in the portfolio})$$

$Floor(T)$: pre-specified safety level to be protected at maturity T (usually 5-10 years)

$Floor(t)$: PV of $Floor(T)$ at $t = w_s(t)\Pi_{safe}(t)$

Cushion: $CPPI(t) - Floor(t)$

Multiplier m: amount leverage (usually 3-6)

$t \to t + \delta t$:

$$CPPI(t+\delta t) = CPPI(t)\left(1 + RAL(t)\left(\frac{\Pi_{risky}(t+\delta t)}{\Pi_{risky}(t)} - 1\right) + (1 - RAL(t))\left(\frac{\Pi_{safe}(t+\delta t)}{\Pi_{safe}(t)} - 1\right)\right)$$

Update RAL as risky portfolio changes.

If $CPPI(t) \le Floor(t)$ → move all holding to Π_{safe}, to be redeemed at maturity

→ investment is (almost) protected at the Floor level

"almost" because of the "gap risk", i.e., sudden drop of the value of Π_{risky} so that there is little time to adjust RAL

Multiplier m:

- amount one can be leveraged for risky asset return
- amount one can tolerate for gap risk

Example:

Total investment amount = \$5m. Amount to be protected: \$4.5m (90%).

Gap risk tolerance: 20% sudden drop in risky asset → m = 1/0.2 = 5

There are numerous variations in CPPI

- Max/min requirements on RAL
- Locking in RAL gain – "ratcheting"
- Dynamical gearing – m as function of volatility level and risky asset performance
- Long date/perpetual CPPI
- Actively/passively managed CPPI
- options on CPPI
- …

Volatility control:

- Volatility targeting
    - invest in different asset classes with a specific portfolio volatility target
    - can be leveraged if market volatility and correlations are low
    - may reinforce market downturn
- Volatility cap

→ High quality volatility forecasting models are essential

Structured notes:

As discussed in previous lectures, there are a large number of varieties of structured products tailored to investors' (and speculators') desired market exposure and risk tolerance.